SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	14-08-2026	**Duration:**	60 minutes

Code of Conduct
I Y.HEMA SAI LAHARI(192311446)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student:Y.HEMA SAI LAHARI

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format:
     As a <user>, I want <feature>, so that <benefit>.
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	**Mark**
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

TEMPLE DONATION AND EVENT MANAGEMENT SYSTEM

A Sprint Planning & User Story Workshop Report

Submitted by

Y. HEMA SAI LAHARI

Roll Number: 192311446

Applying Agile Software Engineering Practices

Academic Year 2025–2026

0. Problem Statement Overview

The Temple Donation and Event Management System is a web and mobile-based platform designed to digitize the operations of a temple trust. The system enables devotees to make online and offline donations, book seva/pooja slots, register for temple events and festivals, receive digital receipts, and stay informed through notifications. On the administrative side, it enables temple management staff, accountants, and priests to manage donations, generate financial and tax reports, plan events, allocate volunteers, and maintain transparent records for trustees and government audit bodies. The current process at most temples is largely manual — donations are recorded in physical registers, receipts are handwritten, and event coordination happens through phone calls — leading to errors, delays, poor transparency, and difficulty in generating accurate reports for the temple trust board and tax authorities.

This document applies Agile Software Engineering practices — specifically Scrum-based Sprint Planning and a User Story Workshop — to translate this problem statement into an actionable, prioritized, and estimated plan for the first development sprint.

1. Project Vision and Stakeholders

1.1 Project Vision Statement

“To build a secure, transparent, and user-friendly digital platform that simplifies temple donation collection, seva/event bookings, and administrative reporting — enabling devotees to contribute and participate with ease, while empowering temple management with real-time financial visibility, accountability, and operational efficiency.”

1.2 Project Objectives

- Provide devotees a simple online channel to donate funds (one-time, recurring, or in-kind) to the temple with instant digital receipts.
- Enable online booking of sevas, poojas, and darshan slots with automated confirmation and reminders.
- Digitize event and festival management, including volunteer allocation and attendee registration.
- Give temple administrators a real-time dashboard for donation tracking, fund utilization, and audit-ready reports.
- Ensure statutory compliance by generating 80G tax-exemption receipts and reports for trust auditors.
- Improve devotee engagement through notifications for upcoming events, festivals, and donation acknowledgements.
- Maintain a secure, role-based system that protects devotee data and financial records.

1.3 Key Stakeholders

Stakeholder	Role / Interest in the System
Devotees / Donors	End users who donate funds, book sevas, register for events, and receive receipts and notifications.
Temple Trust Board / Trustees	Owns the vision, approves major features, reviews high-level financial and audit reports.
Temple Administrator	Manages day-to-day operations: approves events, oversees staff, monitors dashboards.
Accountant / Finance Officer	Records and reconciles donations, generates tax receipts and financial statements.
Priests / Seva Coordinators	Manage seva/pooja schedules, availability, and special event rituals.
Volunteers	Assist in event execution, crowd management, and registration desks; need task assignments.
System Administrator (IT)	Manages user roles, system configuration, backups, and security.
Government / Audit Authorities	External stakeholders requiring statutory compliance reports (e.g., 80G, trust audits).
Development Team (Scrum Team)	Designs, builds, tests, and delivers the product incrementally through sprints.
Product Owner	Owns and prioritizes the product backlog on behalf of the temple trust.

2. Epics and User Stories

Based on the problem statement, the requirements have been grouped into six Epics. Each Epic is broken down into well-structured User Stories following the standard format: “As a <user>, I want <feature>, so that <benefit>.”

EPIC 1: Devotee Donation Management

Enables devotees to make, track, and receive proof of donations through multiple channels.

- US-01: As a devotee, I want to make an online donation using UPI/card/net-banking, so that I can contribute to the temple without visiting in person.
- US-02: As a devotee, I want to receive an instant digital receipt after donating, so that I have proof of my contribution for tax purposes.
- US-03: As a devotee, I want to set up a recurring monthly donation, so that I can support the temple regularly without repeating the process.
- US-04: As a devotee, I want to view my donation history, so that I can track my contributions over time.
- US-05: As a devotee, I want to donate in-kind items (e.g., grains, clothes) by scheduling a drop-off, so that I can contribute beyond monetary donations.

EPIC 2: Seva / Pooja Booking

Allows devotees to browse and book sevas or poojas at the temple.

- US-06: As a devotee, I want to view the list of available sevas with timings and prices, so that I can choose the right seva for my needs.
- US-07: As a devotee, I want to book a seva slot online and pay in advance, so that I don't have to wait in long queues at the temple.
- US-08: As a devotee, I want to receive a confirmation and reminder notification before my booked seva, so that I don't miss my slot.
- US-09: As a priest/seva coordinator, I want to view the day's seva bookings, so that I can prepare and schedule rituals accordingly.

EPIC 3: Event and Festival Management

Covers creation, promotion, and registration for temple events and festivals.

- US-10: As a temple administrator, I want to create and publish a new festival/event with date, description, and capacity, so that devotees can be informed and register.
- US-11: As a devotee, I want to register for an upcoming festival or event, so that I can participate and receive updates.
- US-12: As a temple administrator, I want to assign volunteers to specific tasks for an event, so that the event runs smoothly.
- US-13: As a volunteer, I want to view my assigned tasks and shift timing, so that I know my responsibilities during the event.

EPIC 4: Financial Reporting and Compliance

Provides accountants and trustees with accurate, auditable financial visibility.

- US-14: As an accountant, I want to view a real-time dashboard of total donations collected by date/category, so that I can monitor fund inflow accurately.
- US-15: As an accountant, I want to generate 80G tax-exemption reports for a selected period, so that I can provide statutory compliance documents to auditors.
- US-16: As a trustee, I want to view a summarized monthly financial report, so that I can review the temple's financial health at board meetings.
- US-17: As an accountant, I want to reconcile online and offline (cash/cheque) donations in a single ledger, so that records remain consistent and error-free.

EPIC 5: Notifications and Devotee Engagement

Keeps devotees informed and engaged through automated communication.

- US-18: As a devotee, I want to receive SMS/email/app notifications about upcoming festivals, so that I can plan my visit in advance.
- US-19: As a devotee, I want to receive a thank-you acknowledgement after donating, so that I feel appreciated and trust the platform.
- US-20: As a temple administrator, I want to send bulk announcements to registered devotees, so that important updates reach everyone quickly.

EPIC 6: User Accounts, Roles and Security

Ensures secure, role-based access for all types of users.

- US-21: As a new user, I want to register and log in securely using my mobile number/email and OTP, so that my account and data remain protected.
- US-22: As a system administrator, I want to assign role-based access (devotee, priest, accountant, admin), so that users only see features relevant to their role.
- US-23: As a system administrator, I want all financial and personal data to be encrypted and backed up, so that devotee and temple data remains safe from loss or breach.

3. Acceptance Criteria

Acceptance criteria for the highest-priority user stories (selected for Sprint 1) are defined below using the Given–When–Then format, ensuring each story has clear, measurable conditions of satisfaction.

US-01: Online Donation Payment

- Given a devotee is on the donation page, When they enter an amount and select a valid payment method, Then the payment gateway should process the transaction and confirm success or failure within 10 seconds.
- Given a payment fails or times out, When the transaction is not completed, Then the system must display an error message and must not deduct funds without recording a successful donation.

US-02: Instant Digital Receipt

- Given a donation payment is successful, When the transaction is confirmed, Then a digital receipt with donor name, amount, date, and unique receipt number must be generated and emailed within 2 minutes.
- Given a devotee wants a physical proof, When they tap 'Download Receipt', Then a PDF receipt should be generated matching the emailed copy.

US-03: Recurring Monthly Donation

- Given a devotee sets up a recurring donation, When the scheduled date arrives each month, Then the system must automatically charge the saved payment method and notify the devotee of success or failure.
- Given a devotee wants to stop recurring donations, When they select 'Cancel Subscription', Then no further charges should occur after the current cycle.

US-07: Seva Slot Booking with Advance Payment

- Given a devotee selects an available seva slot, When they confirm and complete payment, Then the slot must be marked as booked and become unavailable to other devotees.
- Given two devotees attempt to book the same slot simultaneously, When one payment completes first, Then the second devotee must be shown 'Slot no longer available' before payment is charged.

US-10: Create and Publish Event/Festival

- Given a temple administrator fills the event creation form with valid date, description and capacity, When they click 'Publish', Then the event must appear in the public events list within 1 minute.
- Given the administrator sets a registration capacity, When the number of registrations reaches that capacity, Then the system must automatically close further registrations and display 'Event Full'.

US-11: Devotee Event Registration

- Given an event is open for registration, When a devotee submits the registration form, Then a confirmation with a unique registration ID must be generated and sent to the devotee.

US-14: Real-Time Donation Dashboard

- Given donations are made through the platform, When an accountant opens the dashboard, Then totals by date, category, and payment mode must be visible and refresh within 5 seconds of a new transaction.

US-15: 80G Tax-Exemption Report Generation

- Given an accountant selects a date range, When they click ‘Generate 80G Report’, Then a downloadable report listing donor name, PAN (if provided), amount, and date must be produced in PDF/Excel format.

US-21: Secure Registration and OTP Login

- Given a new user enters their mobile number, When they request an OTP, Then a 6-digit OTP valid for 5 minutes must be sent, and the account should only be created after successful OTP verification.
- Given a user enters an incorrect OTP three times, When the third attempt fails, Then the system must lock further attempts for 15 minutes to prevent brute-force access.

US-22: Role-Based Access Control

- Given a system administrator assigns the ‘Accountant’ role to a user, When that user logs in, Then only finance-related modules (dashboard, reports, ledger) should be visible, and admin/event-creation modules must remain hidden.

4. Product Backlog Prioritization

The product backlog has been prioritized using a combination of MoSCoW analysis and Business Value scoring (1–10 scale), considering devotee impact, revenue/compliance criticality, and technical dependency. Higher business value and lower dependency generally result in higher priority for early sprints.

ID	User Story (Short)	Business Value (1-10)	MoSCoW	Priority Rank
US-01	Online donation payment	10	Must Have	1
US-02	Instant digital receipt	9	Must Have	2
US-21	Secure registration & OTP login	10	Must Have	3
US-22	Role-based access control	8	Must Have	4
US-07	Seva slot booking + payment	8	Must Have	5
US-06	View available sevas	7	Must Have	6
US-14	Real-time donation dashboard	8	Should Have	7
US-10	Create & publish event/festival	7	Should Have	8
US-11	Devotee event registration	7	Should Have	9
US-08	Seva booking confirmation/reminder	6	Should Have	10
US-15	80G tax-exemption report	8	Should Have	11
US-17	Reconcile online/offline donations	6	Should Have	12
US-04	View donation history	5	Should Have	13
US-19	Thank-you acknowledgement	5	Could Have	14
US-18	Festival notifications	5	Could Have	15
US-09	Priest view of daily bookings	5	Could Have	16
US-12	Assign volunteers to tasks	4	Could Have	17
US-13	Volunteer task view	4	Could Have	18
US-16	Monthly financial summary	6	Could Have	19
US-20	Bulk announcements	4	Could Have	20
US-03	Recurring monthly donation	6	Won't Have (this release)	21
US-05	In-kind donation scheduling	3	Won't Have (this release)	22

ID	User Story (Short)	Business Value (1-10)	MoSCoW	Priority Rank
US-23	Data encryption & backup	9	Must Have (NFR)	—*

**US-23 (data encryption & backup) is a non-functional requirement treated as a cross-cutting 'Must Have' implemented alongside every sprint rather than ranked as a standalone feature.*

Prioritization Rationale:

- Must Have items (US-01, US-02, US-21, US-22, US-07, US-06) form the core transactional flow — without secure login and payment, no other feature can be meaningfully used or tested.
- Should Have items add high business value (financial visibility, event participation) but depend on the Must Have foundation being in place first.
- Could Have items enhance engagement and volunteer coordination but do not block the core donation/booking flow.
- Won't Have (this release) items are valuable but deferred to a later release due to lower urgency and higher implementation complexity relative to value.

5. Sprint Backlog Plan (Sprint 1)

Sprint 1 is a 2-week sprint. The following high-priority user stories were pulled from the top of the product backlog and decomposed into actionable development tasks for the Scrum team.

5.1 Selected Sprint 1 User Stories

- US-01: Online donation payment
- US-02: Instant digital receipt
- US-21: Secure registration & OTP login
- US-22: Role-based access control
- US-06: View available sevas
- US-07: Seva slot booking with advance payment

5.2 Sprint Backlog Task Breakdown

User Story	Development Tasks
US-21: Secure registration & OTP login	1. Design registration/login UI 2. Build OTP generation & SMS/email gateway integration 3. Implement OTP verification & account creation API 4. Add rate-limiting/lockout logic 5. Unit + integration testing
US-22: Role-based access control	1. Design role/permission schema 2. Build admin UI to assign roles 3. Implement middleware to restrict module access by role 4. Test each role's visible modules
US-01: Online donation payment	1. Integrate payment gateway (UPI/card/net-banking) 2. Build donation entry form & amount validation 3. Implement transaction status handling (success/fail/timeout) 4. Log transactions to database 5. Test with sandbox payment credentials
US-02: Instant digital receipt	1. Design receipt template (PDF) 2. Build receipt generation service triggered on successful payment 3. Integrate email service to send receipt 4. Add ‘Download Receipt’ option in devotee portal 5. Test receipt accuracy & delivery time
US-06: View available sevas	1. Design seva catalogue data model 2. Build admin UI to add/edit sevas (name, timing, price) 3. Build devotee-facing seva listing page 4. Test listing accuracy and filters
US-07: Seva slot booking with advance payment	1. Build slot availability & locking logic (handle concurrency) 2. Integrate payment flow with slot booking 3. Build booking confirmation screen 4. Test concurrent booking edge cases

5.3 Sprint Timeline Overview

Day	Activity
Day 1	Sprint Planning Meeting; finalize backlog & task assignment
Day 2–4	Development: authentication (US-21) and role-based access (US-22)
Day 5–7	Development: donation payment (US-01) and receipt generation (US-02)

Day	Activity
Day 8–9	Development: seva catalogue (US-06) and slot booking (US-07)
Day 10	Daily stand-ups throughout; integration testing begins
Day 11–12	Bug fixing, QA testing, UAT with sample devotee users
Day 13	Sprint Review / Demo to Product Owner & stakeholders
Day 14	Sprint Retrospective; backlog refinement for Sprint 2

6. Story Point Estimation

Story points were assigned using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13), where the team estimated relative effort, complexity, and uncertainty for each story rather than exact hours. A story rated as a well-understood, simple 3-point baseline was 'US-06: View available sevas' (a straightforward CRUD + listing feature).

User Story	Story Points	Justification
US-21: Secure registration & OTP login	8	Involves third-party SMS/email gateway integration, OTP generation/expiry logic, security hardening (rate limiting), and thorough edge-case testing — moderate-high complexity and some external-dependency uncertainty.
US-22: Role-based access control	5	Requires a permissions data model and UI, but is a well-understood pattern with limited external dependencies.
US-01: Online donation payment	8	Payment gateway integration carries external-API uncertainty, must handle failure/timeout/retry scenarios reliably, and requires strict security and testing — higher risk and effort.
US-02: Instant digital receipt	5	Depends on US-01 completing; requires PDF generation and email integration, moderate complexity.
US-06: View available sevas	3	Simple CRUD and listing feature with minimal business logic; used as the team's estimation baseline.
US-07: Seva slot booking with advance payment	8	Requires concurrency handling (locking a slot to avoid double-booking) combined with the payment flow — higher complexity and risk of edge-case bugs.

Total Sprint 1 Story Points: 37

Estimation Technique Notes: The team conducted Planning Poker in a workshop setting — each member privately estimated points for a story, revealed simultaneously, and discussed outliers until consensus was reached. Stories involving third-party integrations (payment gateway, OTP/SMS) or concurrency handling (slot locking) were consistently estimated higher due to integration and testing uncertainty, while simple listing/CRUD features were estimated lower.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"By the end of Sprint 1, devotees should be able to securely register/log in, view available sevas, book and pay for a seva slot, and make a donation while instantly receiving a digital receipt — establishing the secure, transactional core of the Temple Donation and Event Management System."

7.2 Expected Sprint Deliverables

- A working devotee registration and OTP-based login module with role-based access control for devotees, priests, accountants, and administrators.
- A functional online donation module integrated with a payment gateway, supporting UPI/card/net-banking payments.
- Automatic generation and email/PDF delivery of digital donation receipts immediately after successful payment.
- A devotee-facing seva catalogue page displaying available sevas with timing and pricing.
- A seva slot booking feature with advance payment and concurrency-safe slot locking to prevent double booking.
- Encrypted storage of user and transaction data, satisfying the baseline security non-functional requirement (US-23).
- A demonstrable, integrated build presented at the Sprint Review, along with updated product backlog and retrospective notes for Sprint 2 planning.

7.3 Definition of Done (Sprint 1)

- All selected user stories meet their acceptance criteria and pass QA testing.
- Code is reviewed, merged, and deployed to a staging environment.
- No critical or high-severity bugs remain open.
- Payment and OTP flows are tested against sandbox/test credentials with documented results.
- Product Owner has reviewed and accepted the sprint demo.

8. Conclusion

This Sprint Planning and User Story Workshop translated the Temple Donation and Event Management System problem statement into a structured, Agile-driven execution plan. Starting from a clear project vision and stakeholder map, the requirements were decomposed into six Epics and twenty-three User Stories, each supported by measurable Given–When–Then acceptance criteria for the highest-priority items. The product backlog was prioritized using MoSCoW and business-value scoring, and the top six stories — covering secure authentication, role-based access, donation payments, digital receipts, and seva booking — were pulled into Sprint 1 with a total estimated effort of 37 story points using Fibonacci-based Planning Poker.

This approach ensures that the very first working increment of the system already delivers real value to devotees (secure login, donations, seva booking) and to temple administrators (secure, auditable transaction records) — laying a solid, transparent, and extensible foundation for subsequent sprints covering event management, financial reporting, and devotee engagement features.

— End of Report —

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